Leo Marcinkus

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Electrical engineering student at CU Boulder with hands-on experience in signal and power integrity, VLSI, PCB design and measurement seeking hardware design and validation work.

EXPERIENCE

Siemens EDA, Software Test Intern (HyperLynx QA) *Boulder, CO | May 2026 - Present*

- Execute and validate HyperLynx SI, PI, AMS, DRC, and Advanced Solver workflows, including DDR4 and DDR5 timing margin, eye diagram, and crosstalk analysis with IBIS models.
- Run power integrity simulations covering **DC drop, target impedance, decoupling, and VRM behavior**, and extract RLGC, current density, and EMI results with advanced solvers.
- Document reproducible software defects and workflow issues in engineering reports and usability recommendations.

Research Assistant, Photonic Neuromorphic Systems (Prof. Wagner) *CU Boulder | May 2026 - Present*

- Contributing to an in-pixel CMOS phase detector and liquid-crystal driver for a 25 x 25 µm pixel on the X-FAB XS018 180 nm image-sensor process.
- **Designing the in-pixel 1.8 V to 3.3 V level shifter** and latch that holds each pixel's comparator decision for a full 1 ms frame and drives the 3.3 V XOR liquid-crystal pad, balancing write margin, leakage, and pixel area.
- Set up the Siemens EDA custom IC flow (Custom IC, Calibre DRC/LVS) and MGLS licensing on a Rocky Linux VM, and evaluate AI-assisted design studies against published work on comparator offset, kT/C and 1/f noise, and charge injection.

Independent Study, RF Receiver Design (Prof. Bogatin) *CU Boulder | January - May 2026*

- Designed a digitally tuned AM receiver front end using a 7-bit switched capacitor bank (2 pF to 302 pF, 128 combinations) on an LC tank, with an Arduino and analog switch ICs replacing a mechanical variable capacitor.
- Validated a working prototype with a Digilent AD3, measuring resonance from 297 kHz to 561 kHz, characterizing 154 pF of stray capacitance from a 0 pF baseline sweep, and recovering AM audio to roughly 10 kHz.

PROJECTS

Golden Arduino Board

- Designed and assembled a custom multi-layer ATmega328 board in Altium, interpreting multiple datasheets and benchmarking it against a stock reference board with oscilloscope measurements.
- Measured up to **1.7x lower ground bounce**, **1.9x better on-die supply isolation**, **2.2x better isolation** from external aggressors, and **4.1x lower near-field emissions** than stock, from local decoupling, tighter return paths, and via stitching.

1T-1C DRAM Cell with Cross-Coupled CMOS Sense Amplifier

- **Designed, simulated, and laid out a 13-transistor two-cell DRAM array** in Siemens Tanner S-Edit, T-Spice, and L-Edit on a Generic 250 nm process at 2.5 V, including precharge, equalization, and SAE/SAEB timing.
- Measured a 57 mV pre-latch sense margin against a 50 mV target, confirmed amplification to a 1.5 V resolved differential, and verified destructive read and partial storage node refresh on both cells.

LABORATORY AND COURSEWORK EXPERIENCE

Graduate High-Speed Digital Design (2026): transmission lines, eye diagrams, PDN and decoupling, S-parameters, SerDes
Graduate High-Speed Digital Measurements (2026): scope and TDR characterization of transmission lines, Dk extraction, 2-port VNA S-parameters, measurement-to-simulation correlation
Graduate High-Speed Digital Design using Ansys HFSS (2026): full-wave 3D field modeling of PCB interconnects and vias
Graduate Electromagnetics for High-Speed Digital Design (2026): L and C matrices, microstrip and stripline field models, PCB waveguide modes and resonances
Embedded Software and RTOS (2025 - 2026): ARM Cortex-M firmware in C, FreeRTOS tasks and mutexes, tracing and profiling
PCB Design and Manufacturing (2025): multi-layer Altium PCBs, signal integrity, crosstalk, and power-rail noise measurement

TECHNICAL SKILLS

Hardware, PCB, and IC Design: Cadence, Altium, KiCad, Tanner (S-Edit, T-Spice, L-Edit), Siemens Custom IC, Calibre
Signal and Power Integrity: HyperLynx SI/PI/AMS and Advanced Solvers, ADS, IBIS/IBIS-AMI, S-parameters, eye diagrams
Embedded Systems: STM32 and ARM Cortex-M, FreeRTOS/CMSIS-RTOS2, Arduino, PWM, SPI/I2C/UART, RISC-V
Programming and Tools: C/C++, Python, MATLAB, SystemVerilog, Assembly, Git, Linux, AI-assisted workflows
Test and Measurement: Oscilloscopes, logic analyzers, DMM, waveform generators, TDR, VNA, I-V characterization

EDUCATION

University of Colorado Boulder *Boulder, CO*

B.S. Electrical Engineering, expected May 2027 | Accelerated M.S., expected December 2027 | Dean's List: Fall 2025, Spring 2026
Interests: Audio electronics, mechanical watches, guitar, piano, NBA